

PRACTICAL – 01 OUTPUT:

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ananya_roy@Ananya:~$ nano practical1.c
ananya_roy@Ananya:~$ gcc practical1.c
ananya_roy@Ananya:~$ ./a.out
Enter a Linux command: ls
Parent PID: 567
Child PID: 568
Parent PID: 567
FIFO1      ex1      malloc_asan  practical1.c  skill1.c
FIFO1.c    exec2.c  myfifo      practical2.c  skill2.c
a.out      execex1.c  namedpipe1.c  prodcons.c   skill3.c
a10        execex2.c  open1.c      prodscons.c  sleep.c
a11        execvex3.c  open3.c      program2.c   student_portal
a2.c       f1        ossp         readwr.c     waitex.c
a7         f1.c      permission_lab receiver.c    waitx.c
a8         heyfifo   pipe1.c      sender.c     xyz
a9         linux_task1 pipe2.c      server       z7
abc        linux_task2 pipe3.c      server.c     zombie.c
b1         linux_task3 pipeex1      shm1.c
b2         linux_task4 pipeex1.c    shm2.c
client.c   malloc    pipeex2.c    sigint.c
dirz       malloc.c  pipeex3.c    signal1.c
Child process finished.
```

```
ananya_roy@Ananya:~$ uname
Linux
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ananya_roy@Ananya:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      12
On-line CPU(s) list:         0-11
Vendor ID:                   GenuineIntel
Model name:                  Intel(R) Core(TM) 5 120U
CPU family:                  6
Model:                       186
Thread(s) per core:         2
Core(s) per socket:         6
Socket(s):                   1
Stepping:                    3
BogoMIPS:                    4992.02
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                             r pge mca cmov pat pse36 clflush mmx fxsr sse s
                             se2 ss ht syscall nx pdpe1gb rdtscp lm constant
                             _tsc rep_good nopl xtopology tsc_reliable nonst
                             op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
                             sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                             opcnt tsc_deadline_timer aes xsave avx f16c rdr
                             and hypervisor lahf_lm abm 3dnowprefetch ssbd i
                             brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
                             d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi
                             2 erms invpcid rdseed adx smap clflushopt clwb
                             sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
                             vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo
                             vdiri movdir64b fsrm md_clear serialize ibt flu
                             sh_lld arch_capabilities

Virtualization features:
  Virtualization:            VT-x
  Hypervisor vendor:         Microsoft
  Virtualization type:       full
Caches (sum of all):
  L1d:                       288 KiB (6 instances)
  L1i:                       192 KiB (6 instances)
  L2:                         7.5 MiB (6 instances)
  L3:                        12 MiB (1 instance)
NUMA:
  NUMA node(s):              1
  NUMA node0 CPU(s):         0-11

```

```

ananya_roy@Ananya:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0     2G   0 disk [SWAP]
sdd   8:48   0     1T   0 disk /mnt/wslg/distro
/

```

```

ananya_roy@Ananya:~$ ps
  PID TTY          TIME CMD
  337 pts/0        00:00:00 bash
  614 pts/0        00:00:00 ps

```

```

ananya_roy@Ananya:~$ top
top - 15:08:39 up 5 min, 1 user, load average: 0.05, 0.03, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7750.6 total, 6256.4 free,  542.8 used, 1103.8 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used. 7207.9 avail Mem

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PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
624	ananya_+	20	0	8312	6124	3888	R	50.0	0.1	0:00.10	top
1	root	20	0	24176	15356	11580	S	0.0	0.2	0:00.96	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.04	init-syste+

```

top - 15:10:02 up 6 min, 1 user, load average: 0.06, 0.04, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7750.6 total, 6249.5 free,  539.6 used, 1114.2 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used. 7211.1 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24176	15356	11580	S	0.0	0.2	0:00.96	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.04	init-systemd(Ub
6	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
46	root	19	-1	50396	16832	15552	S	0.0	0.2	0:00.24	systemd-journal
80	systemd+	20	0	22416	14484	11984	S	0.0	0.2	0:00.12	systemd-resolve
90	root	20	0	35360	12324	9236	S	0.0	0.2	0:00.39	systemd-udev
110	root	20	0	2888	2012	1884	S	0.0	0.0	0:00.04	chronyd-starter
111	root	20	0	4472	3148	2900	S	0.0	0.0	0:00.01	cron
112	message+	20	0	8820	5452	4772	S	0.0	0.1	0:00.05	dbus-daemon
149	root	20	0	44096	29784	14640	S	0.0	0.4	0:00.27	networkd-dispat
167	root	20	0	18444	9416	8164	S	0.0	0.1	0:00.10	systemd-logind
169	root	20	0	1792300	15484	12616	S	0.0	0.2	0:00.12	wsl-pro-service
183	syslog	20	0	220548	5760	4504	S	0.0	0.1	0:00.06	rsyslogd
213	_chrony	20	0	23592	11052	7068	S	0.0	0.1	0:00.05	chronyd
233	_chrony	20	0	12096	2404	1760	S	0.0	0.0	0:00.00	chronyd
239	root	20	0	5492	2828	2644	S	0.0	0.0	0:00.01	agetty
240	root	20	0	123460	32616	18004	S	0.0	0.4	0:00.20	unattended-upgr
335	root	20	0	3196	1112	980	S	0.0	0.0	0:00.00	SessionLeader
336	root	20	0	3196	1248	1108	S	0.0	0.0	0:00.03	Relay(337)
337	ananya_+	20	0	6268	5624	3928	S	0.0	0.1	0:00.08	bash
338	root	20	0	8648	5616	4772	S	0.0	0.1	0:00.01	login
383	ananya_+	20	0	22344	13572	10948	S	0.0	0.2	0:00.16	systemd
385	ananya_+	20	0	22912	4088	2088	S	0.0	0.1	0:00.00	(sd-pam)
413	ananya_+	20	0	6268	5556	3928	S	0.0	0.1	0:00.03	bash
624	ananya_+	20	0	8312	6128	3888	R	0.0	0.1	0:00.15	top